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AI-POWERED NOISE POLLUTION MAPPER & ANALYZER FOR URBAN SPACES

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Abstract : Rapid urban growth has significantly increased environmental noise, creating serious concerns for public health and urban sustainability. Conventional monitoring systems often lack real-time capabilities and fail to capture dynamic variations in city soundscapes. This paper proposes an AI-driven noise pollution mapping and analysis system designed for modern urban environments. The system integrates IoT-based acoustic sensors with advanced signal processing and machine learning techniques to capture, analyze, and classify noise sources such as traffic, construction, and human activity. It generates real-time, location-aware noise maps that highlight spatial distribution and temporal variations in noise intensity. Additionally, predictive modeling is used to forecast future noise trends, enabling proactive urban planning and effective noise management. The proposed framework is scalable, cost-efficient, and demonstrates improved accuracy compared to traditional methods. Overall, it provides a smart and data-driven solution for monitoring and reducing urban noise pollution.

Keyword : AI -Based, Noise pollution, IOT Based

I. INTRODUCTION

Urban environments are rapidly expanding due to population growth, industrialization, and increased transportation activities. While this development supports economic progress, it also introduces significant environmental challenges, among which noise pollution is one of the most overlooked yet harmful factors. Continuous exposure to high noise levels has been linked to various health issues, including stress, sleep disturbances, hearing loss, and cardiovascular problems. Despite its serious impact, noise pollution often receives less attention compared to air and water pollution, mainly due to limitations in monitoring and analysis techniques.

Traditional methods of noise assessment rely on manual measurements or stationary monitoring systems that provide limited spatial and temporal coverage. These approaches fail to capture the dynamic and complex nature of urban soundscapes, where noise levels can vary significantly across locations and time. As cities evolve into smart ecosystems, there is a growing need for intelligent systems that can continuously monitor, analyze, and interpret environmental data in real time. Recent advancements in artificial intelligence and data analytics have opened new possibilities for addressing such challenges. Machine learning models can process large volumes of sensory data, identify patterns, and classify different types of noise sources with high accuracy. When combined with geospatial mapping technologies, these capabilities enable the creation of detailed noise maps that reflect real-time conditions across urban areas.

This paper proposes an AI-powered noise pollution mapper and analyzer tailored for urban spaces. The system integrates distributed sensing devices, real-time data processing, and intelligent classification algorithms to generate dynamic and interactive noise maps. Furthermore, it incorporates predictive analysis to estimate future noise trends, supporting proactive decision-making. By offering a scalable and data-driven solution, the proposed approach aims to enhance environmental monitoring and contribute to the development of healthier and more sustainable cities**.**

II. LITERATURE

a) 2015 – Karol Pic zak: ESC-50 Dataset Pic zak's 2015 work is widely recognized as a foundational contribution to environmental sound classification research. By introducing the ESC-50 dataset—consisting of 2,000 carefully labeled audio clips spanning 50 everyday sound classes Pic zak provided the field with a standardized benchmark that had been missing. The dataset covers a broad range of environmental categories, from animal sounds to human activities and natural phenomena.

b) 2017 – Urban GIS Noise Vulnerability Study The 2017 study in urban GIS noise vulnerability applied spatial analytics to understand how noise pollution affects different parts of a city. By integrating measured noise levels with key spatial variables such as land use types, traffic intensity, and building structure distribution. the authors created a model capable of identifying neighbourhoods most at risk from excessive noise exposure. This work demonstrated the value of combining environmental data with GIS techniques to support municipal planning, zoning decisions, and the development of targeted noise mitigation strategies. It also highlighted how noise vulnerability is shaped by the interaction of physical infrastructure and human activity.

c) 2018 – Z. Zhang: Deep CNN with Mix-up for Sound Classification
Zhang's 2018 research showcased the growing potential of deep learning for environmental sound classification. The study proposed a deep convolutional neural network enhanced by the mix-up data-augmentation technique, which blends pairs of audio samples to improve generalization. Evaluated on popular datasets like UrbanSound8K and ESC-50, the model achieved competitive performance, underscoring how combining CNN structures with advanced augmentation can reduce over fitting and improve robustness. This work is frequently cited for demonstrating that deep learning architectures, even without highly handcrafted features, can excel in environmental audio tasks when supported by effective data augmentation.

d) 2022 – Deep Learning-Based ESC Framework (Tech Science CMC)
The 2022 study published in CMC (Tech Science Press) proposed a feature-rich deep learning framework for environmental sound classification. Rather than relying on a single representation, the model integrated multiple audio features—including Mel-spectrograms, MFCCs, and other time–frequency descriptors—to capture different acoustic properties. Through experiments on UrbanSound8K and ESC-50, the authors reported high accuracy, highlighting the advantage of combining complementary feature sets with modern neural architectures. The study reinforced the understanding that environmental sound classification benefits from multi-representation input designs that allow models to learn both detailed spectral patterns and general temporal structures.

e) 2024 – GeoAI Noise-Prone Area Assessment (Tehran Case Study)
In 2024, a GeoAI-based study conducted in Tehran demonstrated how machine learning and geospatial methods can be combined to identify noise-prone areas with high precision. Using spatial predictors such as traffic road networks, land-use types, and proximity to commercial zones, the researchers applied gradient boosting and explainable AI techniques to model noise exposure across the city. The resulting maps offered granular insights into urban noise distribution and revealed the underlying factors contributing to acoustic hotspots. This work exemplifies the growing trend of using interpretable GeoAI models for environmental planning and targeted noise-mitigation policies.

III. PROPOSED METHODOLOGY

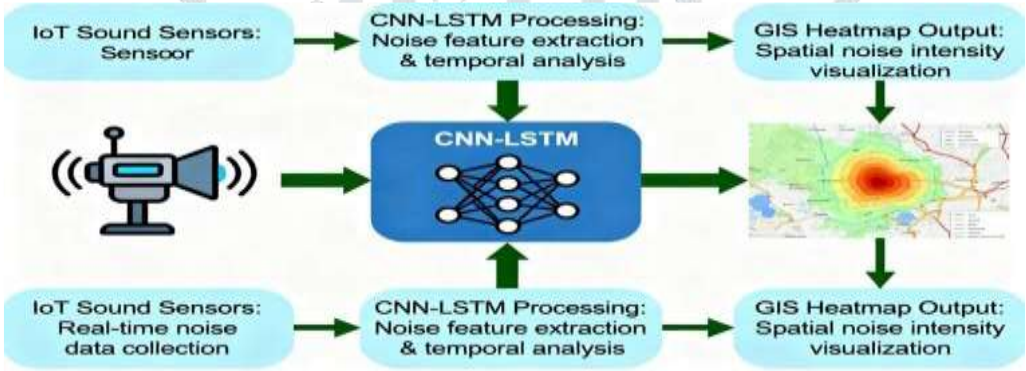
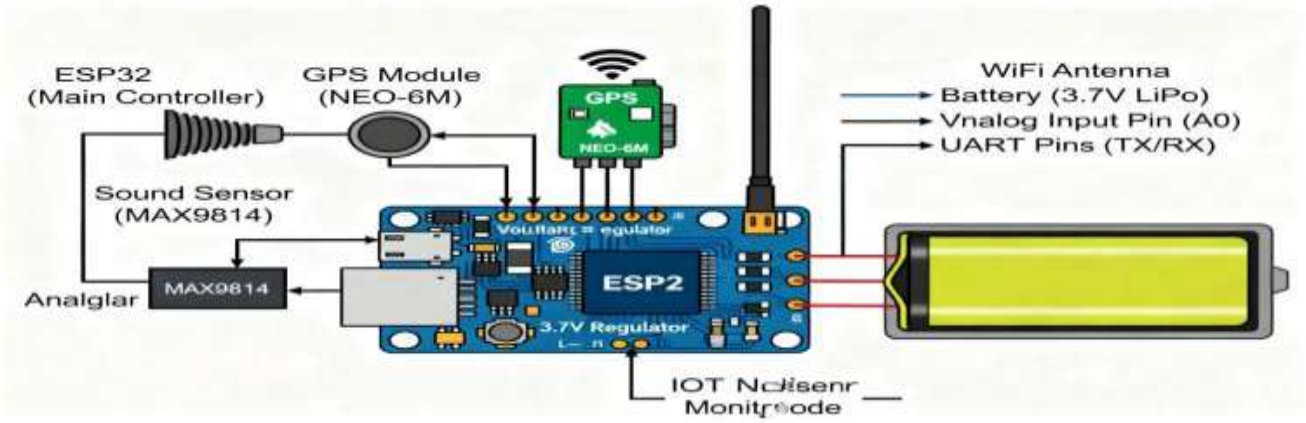


fig 1: workflow: ai-powered noise pollution mapper & analyzer for urban space

This complete workflow of the proposed AI-powered noise pollution mapping and analysis system for urban environments. It begins with IoT sound sensors, which are deployed across different locations in a city to continuously capture real-time noise data. These sensors collect acoustic signals along with time-based information, forming a continuous stream of environmental sound data. The collected data is then passed to the CNN-LSTM processing module, where advanced deep learning techniques are applied. In this stage, the Convolutional Neural Network (CNN) extracts important noise features such as frequency patterns and intensity variations, while the Long Short-Term Memory (LSTM) network analyzes temporal patterns to understand how noise changes over time. This hybrid model enables accurate classification and pattern recognition of different noise sources.
After processing, the analyzed data is sent to the GIS heatmap output module, where it is transformed into spatial visualizations. These heatmaps represent noise intensity across different geographic locations using color gradients, making it easy to identify high-noise zones or hotspots within the city. The diagram also shows a continuous loop, indicating that real-time data collection and analysis are ongoing processes, allowing the system to dynamically update noise maps. Overall, the diagram illustrates an end-to-end pipeline that integrates IoT sensing, AI-based analysis, and geospatial visualization to provide an intelligent and real-time noise monitoring solution for urban planning and environmental management.

Hardware used:



ig 2: circuit diagram: ai-powered noise pollution mapper & analyzer for urban space

An IoT-based noise monitoring node designed to collect and transmit real-time environmental sound data. At the core of the system is the ESP32 microcontroller, which acts as the main processing and communication unit. It is responsible for receiving input signals from sensors, processing the data, and transmitting it wirelessly using its built-in Wi-Fi capability. After acquiring the raw audio and GPS data, the ESP32 processes lightweight filtering operations and transmits the collected information to the server through its built-in Wi-Fi module. In scenarios requiring wider coverage or remote deployment, the ESP32 can interface with external GSM modules for cellular connectivity. Power is supplied by a 3.7V Li-Po battery, regulated to appropriate voltage levels using onboard converters, ensuring long-term autonomous operation in outdoor environments. This hardware configuration, reflected in the circuit diagram, creates a compact, energy-efficient sensing unit capable of continuous noise monitoring.

The sound sensor (MAX9814) is used to capture ambient noise. This module contains a microphone and automatic gain control, which allows it to detect sound levels accurately even in varying environments. The analog output from the MAX9814 is connected to the analog input pin (A0) of the ESP32, where the sound signal is converted into digital data for further processing.

A GPS module (NEO-6M) is integrated into the system to provide precise geographical location information. It communicates with the ESP32 through UART pins (TX/RX), enabling the system to tag each noise reading with latitude and longitude coordinates. This is essential for mapping noise data geographically.

The system is powered by a 3.7V LiPo battery, which supplies energy to the ESP32 and connected components. A voltage regulator ensures stable power delivery, protecting the components from voltage fluctuations. For communication, a Wi-Fi antenna (integrated with ESP32) enables wireless data transmission to a cloud server or central system. All these components together form an IoT noise monitoring node, where the sound sensor captures noise, the ESP32 processes and combines it with GPS data, and then transmits the information over Wi-Fi. This setup enables continuous, real-time, and location-aware noise data collection for further AI-based analysis and visualization.

The proposed system is an intelligent and scalable solution designed to monitor, analyze, and visualize urban noise pollution using a combination of IoT, artificial intelligence, and geospatial technologies. The architecture begins with distributed IoT-based sound sensing nodes, where environmental audio signals are continuously captured along with precise location information. These nodes transmit real-time data to a centralized processing system, forming the foundation for dynamic noise analysis.

Software implementation

At the core of the system lies an AI-driven analytical engine developed using the Python programming language. Python is utilized for handling data pipelines, preprocessing audio signals, and implementing machine learning workflows. The incoming audio data is processed using the Librosa library, which extracts essential acoustic features such as Mel-Frequency Cepstral Coefficients (MFCC), Zero Crossing Rate (ZCR), and spectral characteristics. These features provide a compact and meaningful representation of complex sound signals.

For intelligent classification and temporal analysis, the system employs a hybrid deep learning model built using TensorFlow. The Convolutional Neural Network (CNN) component identifies spatial patterns in the audio features, enabling the recognition of different noise sources such as traffic, construction, and human activity. Simultaneously, the Long Short-Term Memory (LSTM) network captures temporal dependencies, allowing the system to understand how noise evolves over time and detect recurring patterns or anomalies. This combination enhances both accuracy and contextual understanding of urban soundscapes.

To enable seamless communication and data handling, a backend server is developed using Flask. Flask manages API requests, processes incoming sensor data, and coordinates interactions between the AI model, database, and visualization layer.

Overall, the system creates a continuous pipeline where raw audio data is collected, transformed into meaningful features, analyzed using deep learning models, and visualized on interactive maps. This integrated approach not only enables real-time monitoring but also supports predictive analysis and data-driven decision-making. By combining IoT sensing, AI intelligence, and GIS visualization, the proposed solution provides an efficient and modern framework for managing noise pollution in smart urban environments.

CONCLUSION

This paper presents an integrated and intelligent framework for monitoring and analyzing urban noise pollution using a combination of IoT, artificial intelligence, and geospatial technologies. The proposed system successfully addresses the limitations of conventional noise monitoring methods by enabling continuous, real-time data collection and automated analysis. Through the use of distributed sound sensors, the system captures dynamic acoustic information, which is then processed using a hybrid CNN-LSTM model to accurately classify noise sources and understand temporal variations.

The incorporation of advanced audio feature extraction techniques and deep learning models enhances the system's ability to interpret complex urban sound environments. Furthermore, the integration of GIS-based visualization provides an intuitive representation of noise distribution through interactive heatmaps, allowing users to easily identify high-noise zones and analyze spatial patterns. The use of cloud-based storage and backend frameworks ensures scalability, reliability, and efficient data management.

Overall, the proposed solution offers a comprehensive, cost-effective, and scalable approach for urban noise monitoring. It not only improves the accuracy and efficiency of noise analysis but also supports predictive insights, enabling proactive decision-making for noise control and urban planning. This system has strong potential to contribute to the development of smarter, healthier, and more sustainable cities by providing actionable data to policymakers and environmental authorities

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